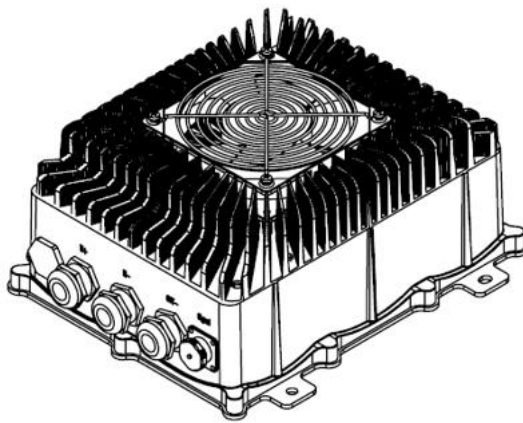

YSTECH bidirectional automatic Buck and Boost DC/DC Converter

100A 2~7Kw Series non isolated DC/DC

User 's manual



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DCFF-UD-100LPB Series Rectifier User's Manual	
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Publication instructions

Content introduction

In order to guide relevant power equipment manufacturers to use our company's relevant products to design and produce power supply systems, this technical guidance is hereby prepared.

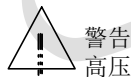
This document describes in detail the appearance, function, parameter index, interface definition and operation description of DCFF-100-LPB series high frequency switch rectifier module.

Audience

This document is suitable for power supply cooperative manufacturers, power users, power maintenance engineers, etc.

Agreement of this document

一、Mark on the product



Affix this mark where high voltage exists.



Add this tag on the protective grounding terminal of the cabinet bottom frame.

二、Mark in manual



Note

The word "attention" refers to the condition or practice that may cause damage to the cost equipment or other equipment.

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1 General

The DCFF-100-LPB series DC/DC converter is a non isolated DC/DC converter with DC voltage input and adjustable DC voltage output. This module adopts DSP digital control, resonant soft switching, active PFC technology, and has multiple patent protections; This converter has automatic buck and boost regulation function, especially with bidirectional energy flow function, which is particularly suitable for connecting battery packs and motor controllers in application scenarios to achieve the function of motor controller energy feedback to battery packs. The converter also has the characteristics of high power density, high power factor, low harmonics, and high efficiency.

2 Main features

- **High efficiency**
Reduce energy consumption, meet the requirements of energy conservation and emission reduction, and save electricity for customers.
- **High power density**
It can save customer space and reduce system cost.
- **DSP digital control**
Fewer devices, higher environmental stability, higher reliability, and more convenient capacity expansion.
- **Low input harmonics**
Reduce the pollution to the power grid and improve the adaptability of the power grid.
- **Output adjustable function**
Adjustable output voltage and output mode function.
- **Wide working temperature range**
Wide operating temperature range to meet most severe working environments
- **hot swapping technology**
It is suitable for convenient use and reduces maintenance costs.
- **Bidirectional energy flow Technology**
Automatic control function for bidirectional energy flow.
- **Built in anti reverse protection, with failure self isolation function**
Greatly improve system reliability.
- **Perfect fault self detection prompt**
Rich fault detection, convenient for customer maintenance.

Product Selection Table

Model	Battery side voltage (V)	DC BUS side voltage (V)	Current range (A)	Power range (kW)
DCFF-UD-100LPB-150	20-130	20-130	0-140	2-7
DCFF-UD-100LPB-200	20-180	20-180	0-140	2-7

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3 Technical indicators

3.1 Technical Parameters

Table 1: Module parameter table

Item	Parameters		
➤ Size & Type			
Size(unit:mm)	285(L)×245(W)×122(H)		
Weight	≤6kg		
Rated Output Power	2~7kW		
Current	0~140A		
DC output Range	DC 20V~180V		
Efficiency	≥97%（Peak value）		
Communication	CANB 2.0		
Model	DCFF-UD-100LPB-100	DCFF-UD-100LPB-150	DCFF-UD-100LPB-200
➤ DC BUS Side (DC Output, Motor controller, lithium battery, DC microgrid, DC bus, etc)			
voltage	20Vdc~70Vdc	20Vdc~110Vdc	20Vdc~150Vdc
Current	0~140A	0~140A	0~140A
Peak to Peak noise	+/- 2.5V		
voltage accuracy	≤±1%		
Current accuracy	≤±1%		
Undervoltage protection	20V Recovery:no self recoverable	20V Recovery:no self recoverable	20V Recovery:no self recoverable
Overvoltage protection	80V Recovery:no self recoverable	120V Recovery:no self recoverable	170V Recovery:no self recoverable
Overcurrent protection	120A Recovery:no self recoverable	125A Recovery:no self recoverable	125A Recovery:no self recoverable
Over temperature protection	90℃ Recovery:80℃	90℃ Recovery:80℃	90℃ Recovery:80℃
External auxiliary power supply activation working	24V or 12V Provide wake-up main DCDC to drive fans, relays, board level power supply, etc. The required power Less than 15W		
Bidirectional control	When the DC bus side of the DC/DC is connected to a motor controller, set the DC BUS Side of the DC/DC to a constant voltage working mode. When a reverse energy feedback occurs from the motor controller, the feedback energy will be automatically reversed back to the battery side of the DC/DC to charge the battery pack.		
➤ Battery Side (DC Input, Supercapacitors, lithium batteries, flow batteries, hydrogen fuel cells, etc)			
Input voltage	20Vdc~70Vdc	20Vdc~110Vdc	20Vdc~150Vdc
Input Current	0~140A		
Peak Current	150A @1 Min		
control model	Battery Side: constant current DC BUS Side: constant voltage		
Undervoltage protection	20V Recovery:no self recoverable	20V Recovery:no self recoverable	20V Recovery:no self recoverable
Overvoltage protection	80V Recovery:no self recoverable	120V Recovery:no self recoverable	170V Recovery:no self recoverable
Overcurrent protection	120A Recovery:no self recoverable	125A Recovery:no self recoverable	125A Recovery:no self recoverable
➤ Working environment conditions			
Working environment temperature	-40℃~60℃ Normal operation; 60℃~80℃ Linear load reduction to 0		

Storage temperature	-40℃～70℃
Relative Humidity	0～95%
Elevation	Full load output below 2000m
➤ Product Safety & Reliability	
Insulation strength	Input to shell, output to housing, input to output, input to COM, output to COM: DC 3500v, 1min; COM to Shell: DC 1400v, 1min;
MTBF	>120000 Hours
EMC emission	Reference standard: IEC 61000-6-3 (Class B)
EMC immunity	Reference standard: IEC 61000-6-1

4 System control

4.1 System diagram

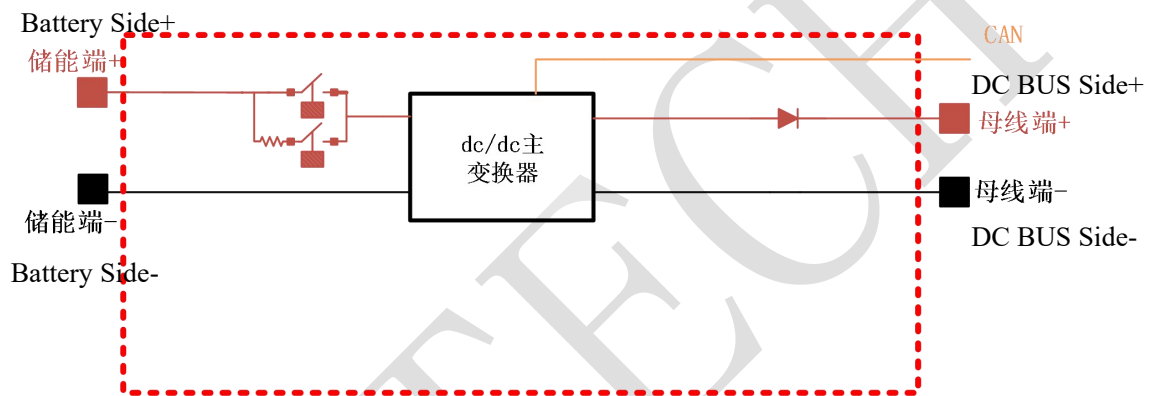


Fig 1 System diagram

4.2 Relay logic

- (1) Energy storage end pre charging enable: the energy storage end pre charging relay is on, the energy storage end main relay is on, and the energy storage end pre charging relay is off; If it fails, the pre charge will report an error,
- (2) Start, discharge, stop,
- (3) Turn off the energy storage terminal relay,
- (4) Close the bus terminal relay.

4.3 constant current and voltage limiting control

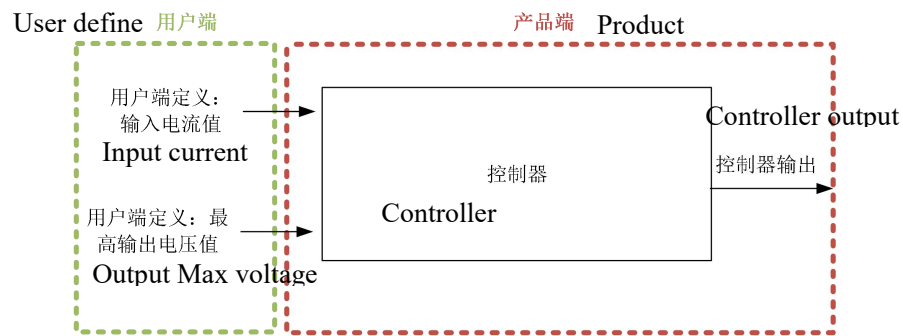


Fig 2 Block diagram of constant current and voltage limiting control

Operating instructions:

1. Energy storage loads (lithium-ion batteries, lead-acid batteries, supercapacitors, etc.): The target current is set to the user's required fuel bus terminal current, and the target voltage value is set to the highest voltage of the energy storage load required by the user.
2. Passive loads (resistors, electrical appliances, etc.): The target current is set to the maximum target value of 300A, and the target voltage value is set to the required voltage of the electrical load. The system will operate in constant voltage mode at the bus end

4.4 System control logic

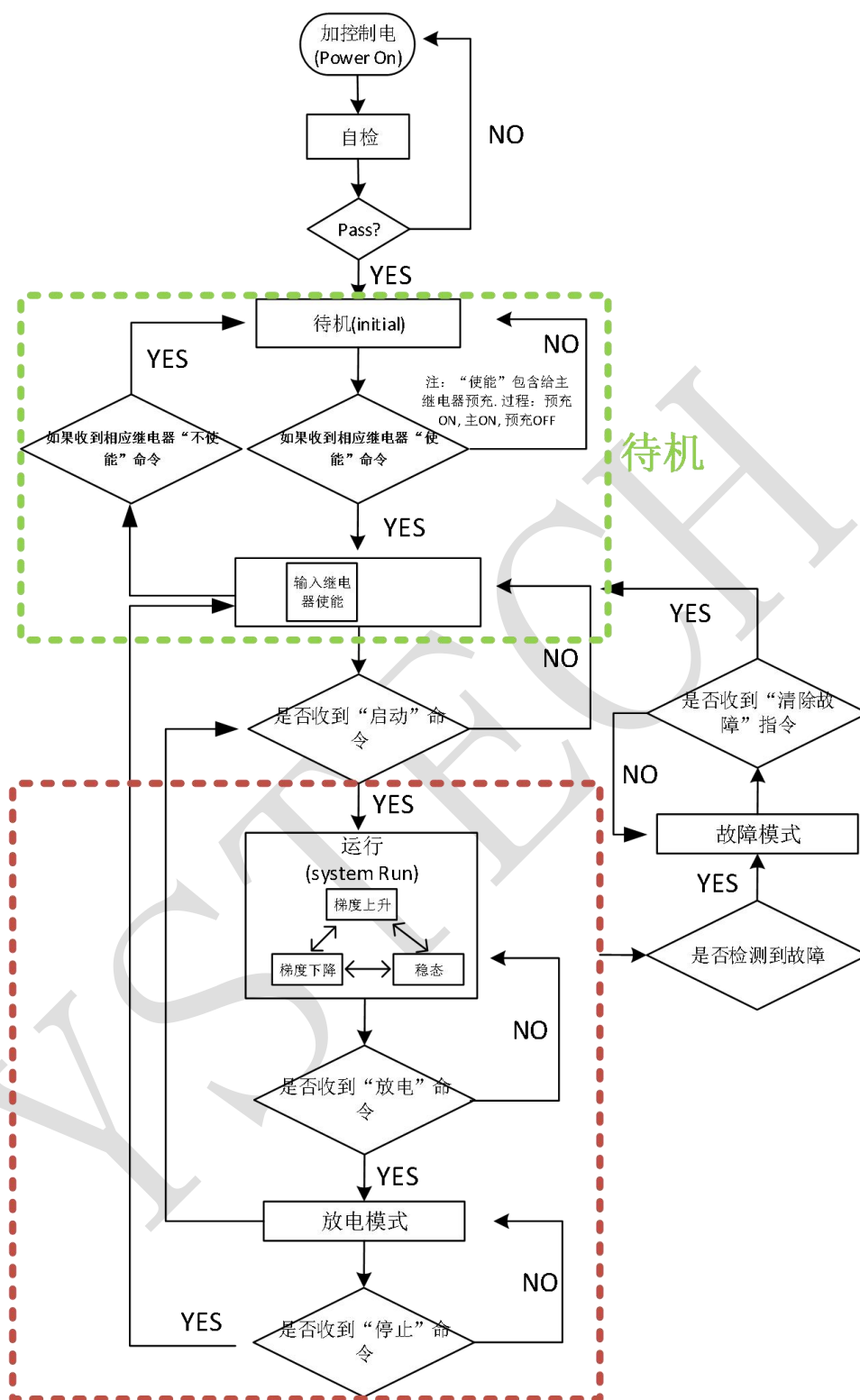


Fig 3 control logic

Note:

Taking the current control mode as an example, the DC/DC converter has four stable states: stop (standby), run (including steady state, gradient rise, gradient drop), discharge, and fault. The transition conditions of each state are switched according to the requirements of the entire vehicle.

DC state: 0 standby; 1. Operation; 2 discharge; 3: Fault

DC command: 0 stop; 1. Start up; 2 discharge; 3: Clear errors

5 Shape structure & Interface

5.1 Shape structure and Size

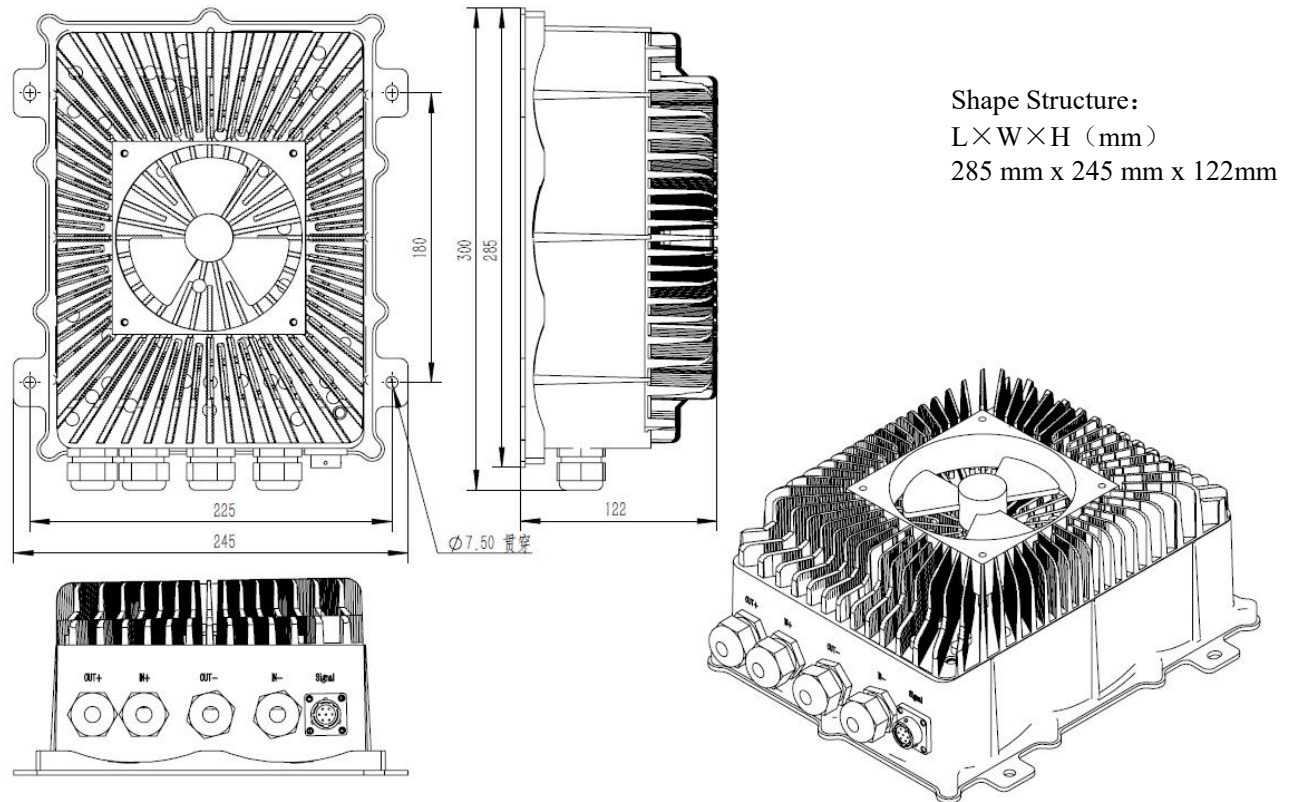


Fig 5.1 over view

5.2 plug pin define

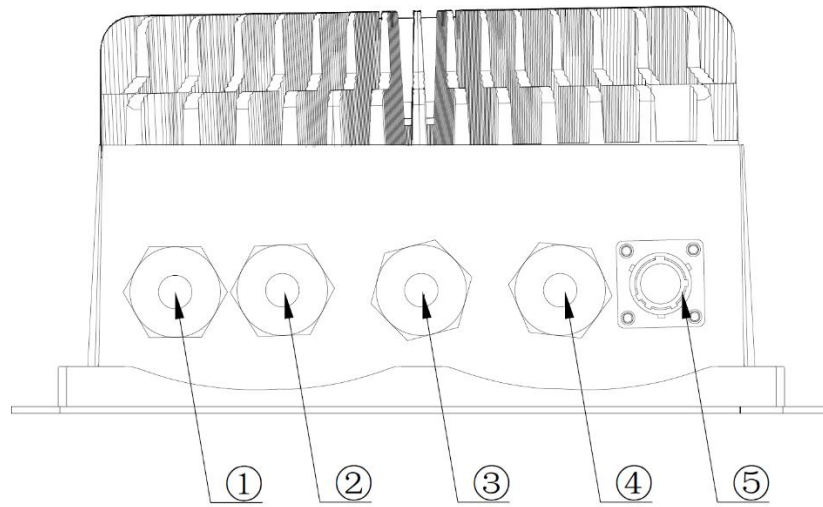
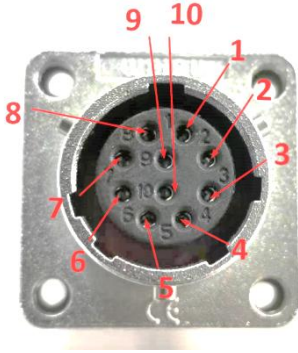


Fig 5.2 pin number

No.	Plug name	Manufacture	Product side model	Cable sid model	Cable ϕ (mm)
1	Positive of DC BUS Side	-	Waterproof cable joint M24*1.5	-	25 mm ²
2	Positive of Battery Side	-	Waterproof cable joint M24*1.5	-	25 mm ²
3	Negative of DC BUS Side	-	Waterproof cable joint M24*1.5	-	25 mm ²
4	Negative of Battery Side	-	Waterproof cable joint M24*1.5	-	25 mm ²
5	Signal plug	威浦	WY16K10Z	WY16J10TE	0.3 mm ²

signal connector

supply	DC/DC side	Electrical terminal connector schematic diagram				
威浦	WY16K10Z					
supply	Cable side					
威浦	WY16J10TE					
Pin No.	Cable wire	function	Signal type	interface	Rated (A)	note
1	CAN-H	通讯线	CAN-H	VCU or other ECU	5	
2	Reserved					
3	Reserved					(12V 控制电+备用)
4	Auxiliary power 24V+	Power supply	24V+	Vehicle low-voltage power supply	5	Auxiliary power supply for activation work provided by external sources Less than 15w
5	Auxiliary power GND	Power supply	GND	Vehicle low-voltage power supply	5	
6	Reserved				5	(12V 控制电-备用)
7	Reserved					
8	CAN-L	Communication cable	CAN-L	VCU or other ECU	5	
9	CAN-GND	Communication cable	CAN Shielding layer	VCU or other ECU	5	
10	Reserved					

6 Packaging, transportation and storage

6.1 Packaging

This equipment is packed in cartons and packed separately. It is packed with shockproof foaming plastic. The packing box also contains the packing list, installation and operation instructions, and each rectifier module is pasted with the factory certificate.

6.2 Transportation

The transportation of rectifier module should be carried out under the condition that the package is in good condition, and there should be no violent vibration and collision during the transportation to prevent moisture and rain.

6.3 Storage

The rectifier module shall be stored in the room with $-40\text{ }^{\circ}\text{C} \sim +75\text{ }^{\circ}\text{C}$, relative humidity not greater than 95%, no corrosive gas and ventilation, and the storage period is 1 year.

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